Data and frequency tables

1 Complete the frequency table below based on the following data set:

$$2, 5, 4, 5, 5, 5, 3, 4, 5, 5, 1, 3, 3, 3, 5, 2, 4, 1, 6, 5, 6, 3, 1, 1, 6$$

Score	Frequency
1	
2	
3	
4	
5	
6	

2 Complete the following frequency table:

Score (x)	Frequency (f)	xf
1	7	
2	4	
3	7	
4	8	
5	8	
6	8	

3 State the mode of the following data set:

Score	Frequency
3	3
4	3
5	4
6	2
7	5
8	8

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4 Find the mean for the following data set, correct to one decimal place:

Score (x)	Frequency (f)	xf
2	7	14
3	2	6
4	8	32
5	5	25
6	4	24
7	7	49

5 A statistician organised a set of data into the following frequency table:

Score (x)	Frequency (f)	xf
5	14	
7	4	
9	2	
11	18	
13	6	
Total		

a Complete the frequency distribution table.

b Find the mean, correct to two decimal places.

6 Consider the data provided in the table:

Score	Frequency
68	16
69	41
70	30
71	31
72	49
73	29

a Find the range.

b Find the mode.

7 Consider the following data set:

$$27, 25, 24, 24, 24, 24, 24, 25, 23, 24, 26, 23, 27,$$
$$23, 24, 27, 23, 23, 27, 25, 24, 24, 27, 25, 23$$

a Construct a frequency distribution table for the data.

b Find the mean, correct to one decimal place.

c Find the range.

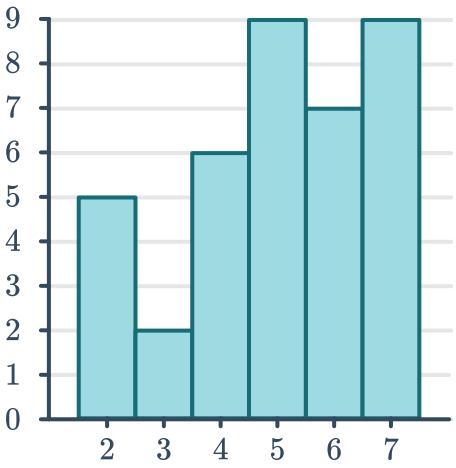
d Find the mode.

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Frequency histograms

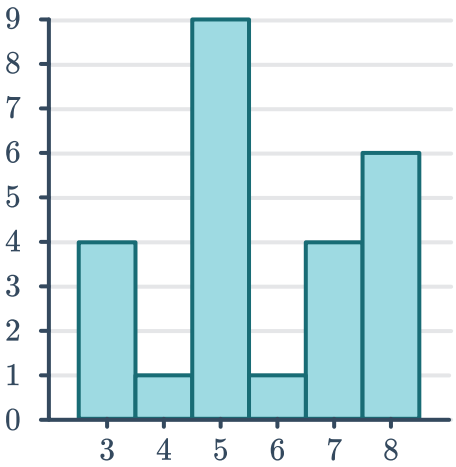
8 Complete the following frequency tables using the given histograms:

a



Score	Frequency
2	
3	
4	
5	
6	
7	

b



Score (x)	Frequency (f)	$f \times x$
3		
4		
5		
6		
7		
8		

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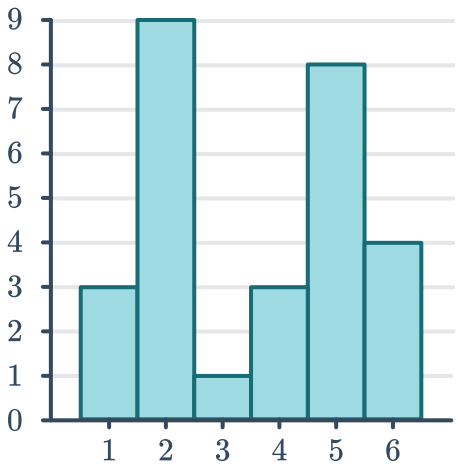
9 For each of the following histograms:

i Find the mode.

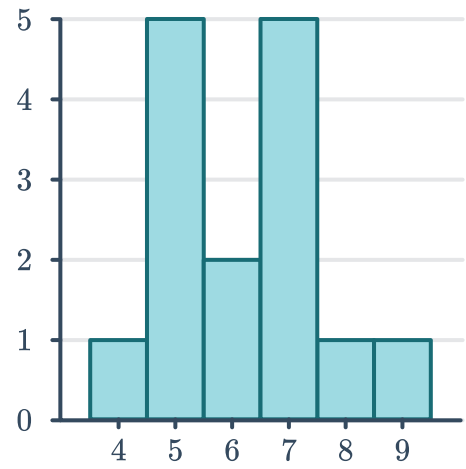
ii Find the median.

iii Find the mean, correct to one decimal place.

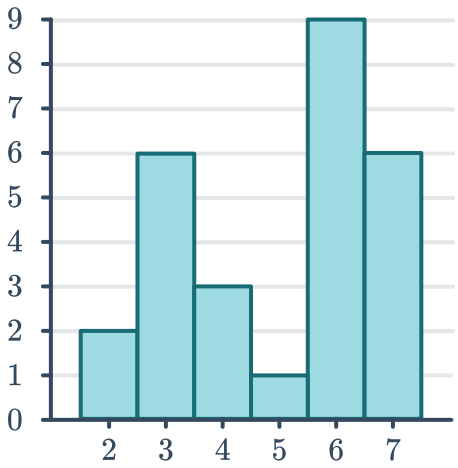
a



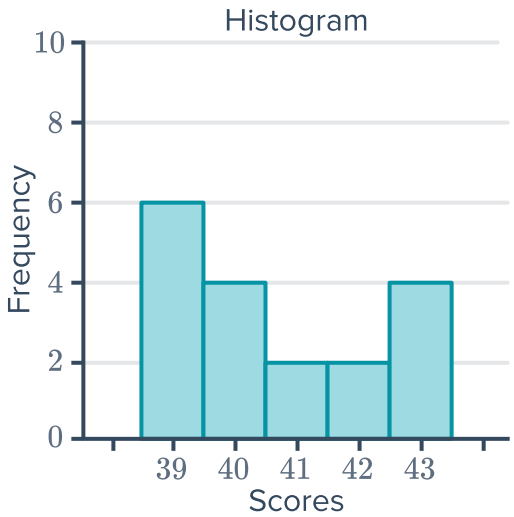
b



c

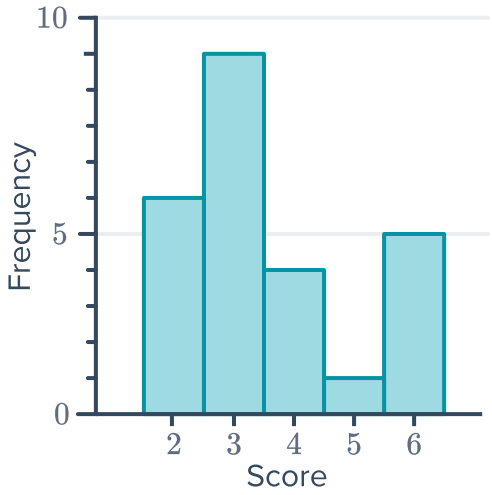


d



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10 Consider the frequency histogram:



Score (x)	Frequency (x)	$f \times x$
2		
3		
4		
5		
6		
Total		

a Complete the frequency distribution table.

b Calculate the mean, correct to one decimal place.

c Find the range of the data.

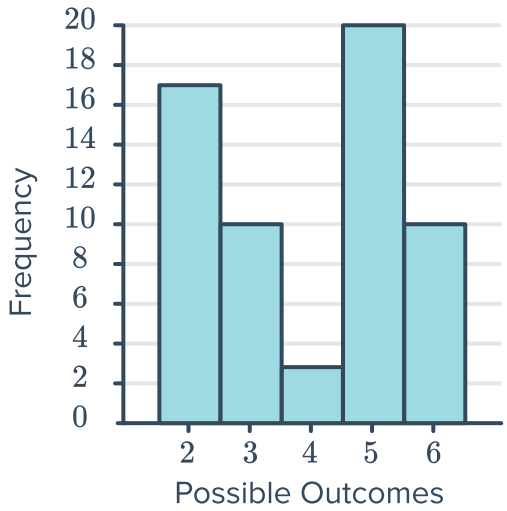
d Find the mode of the data.

11 For each of the following histograms:

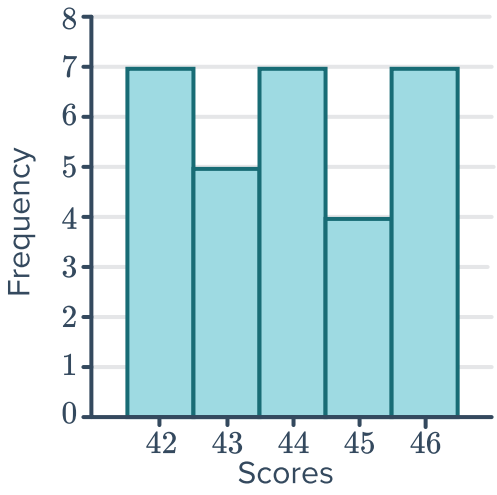
i Calculate the mean, to two decimal places.

ii Find the mode(s).

a



b

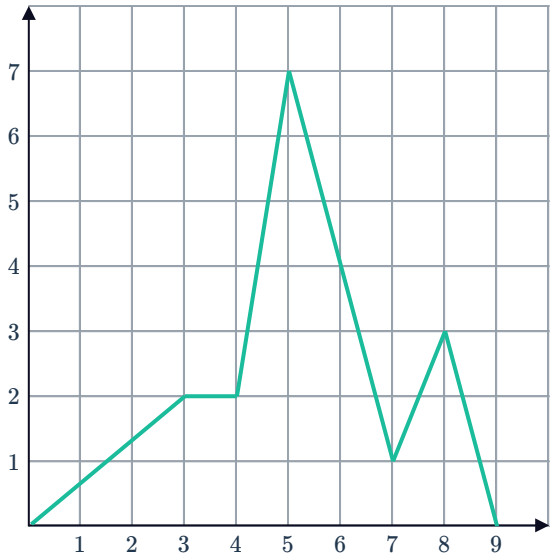


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Frequency polygons

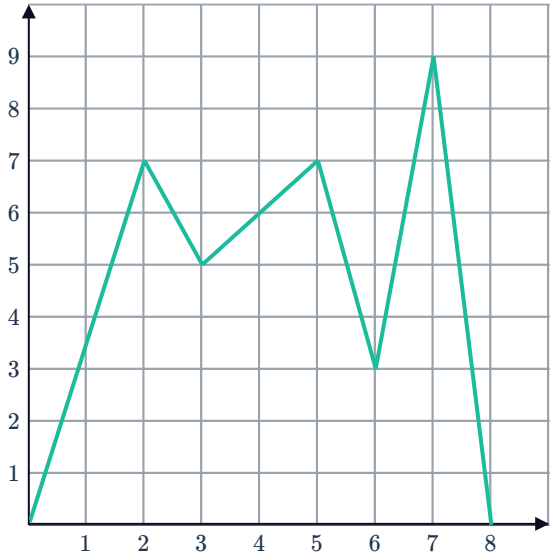
12 Complete the following frequency tables using the given frequency polygons:

a



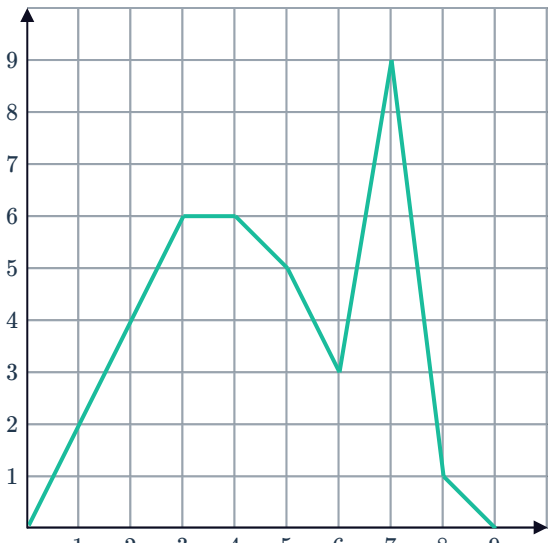
Score	Frequency
3	
4	
5	
6	
7	
8	

b



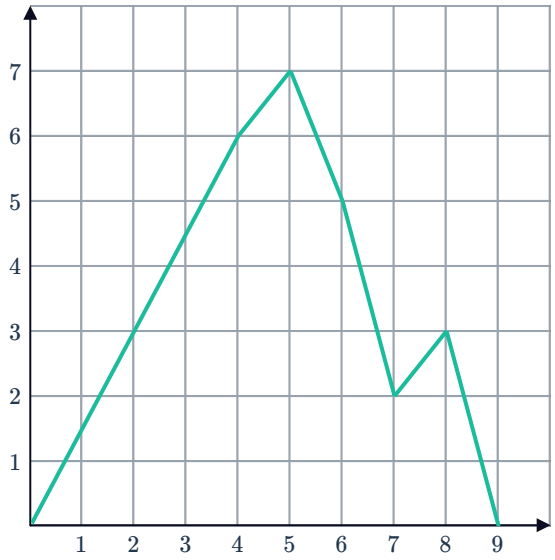
Score (x)	Frequency (f)	$f \times x$
2		
3		
4		
5		
6		
7		

13 Find the mode for the following data set:

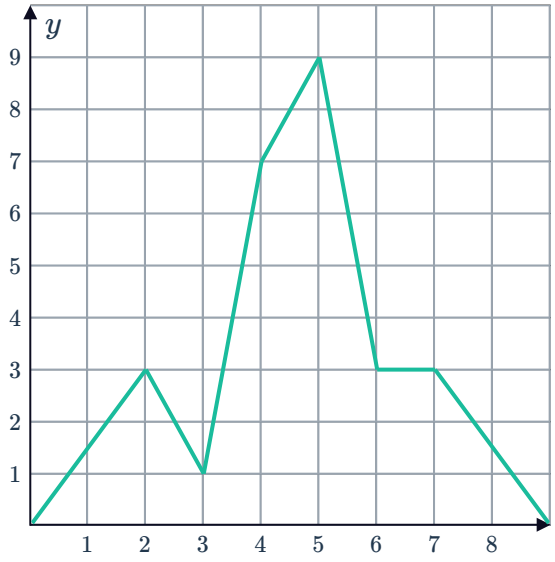


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14 Find the median for the following data set:



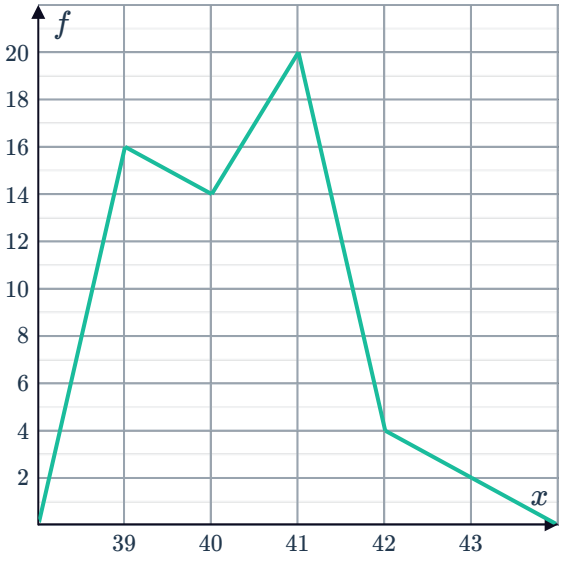
15 Find the mean for the following data set, correct to one decimal place:



16 Consider the frequency polygon shown:

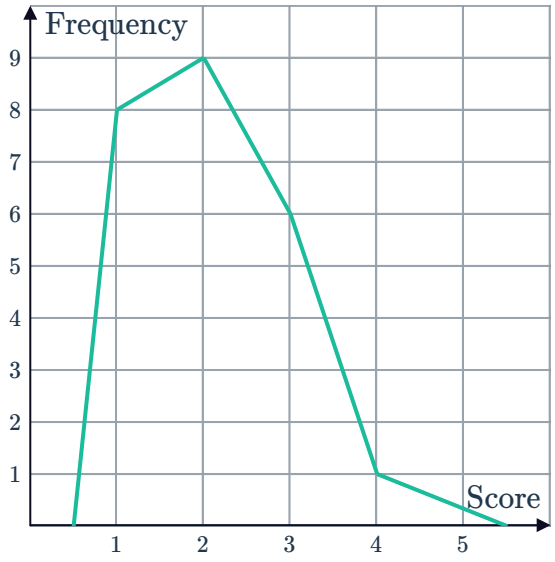
a Calculate the mean of the scores, correct to two decimal places.

b Find the mode.



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17 Consider the following frequency polygon:



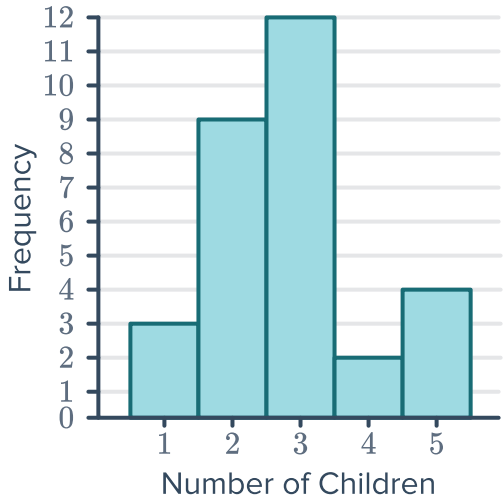
a Calculate the mean.

b Find the median.

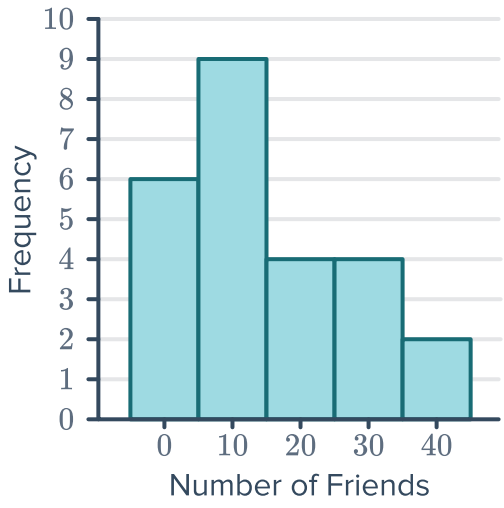
c Find the mode.

Applications

18 A group of families were surveyed on the number of children they have and the results are shown in the column graph:
What is the average number of children in a family from the survey?
Round your answer to one decimal place.

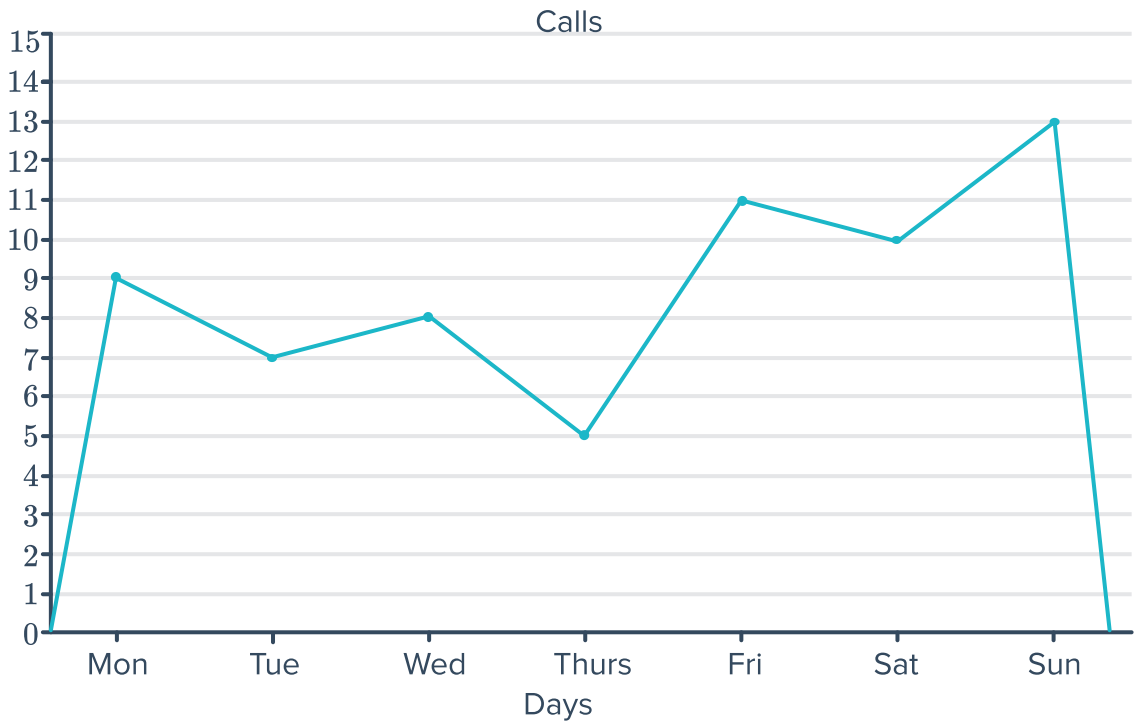


19 Some people were asked approximately how many of their high school friends they remained in contact with after high school. The results are presented in the column graph:
What is the average number of friends that people in the survey remained in contact with?



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20 The frequency polygon shows the frequency of calls made during each day of the week:



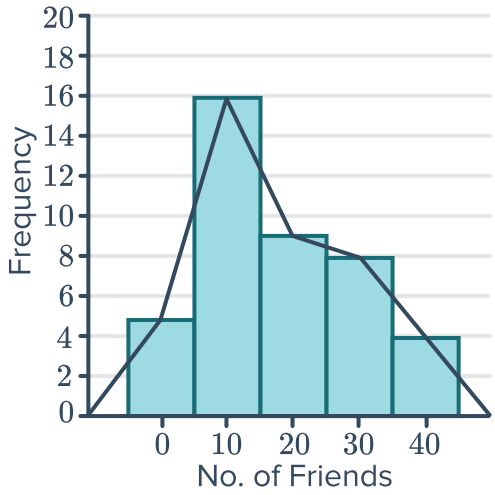
a How many calls were made on Friday?

b What was the maximum number of calls made on a single day?

c What was the total number of calls made during the week?

d Find the mean number of calls made during the week.

21 A group of people were asked approximately how many of their high school friends they remained in contact with after high school to the nearest 10 friends. The results are presented in the following frequency polygon:



a What was the most common response?

b Find the mean number of people who kept in touch with friends after high school.
Round your answer to the nearest whole number.

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22 Mr. Rodriguez recorded the number of pets owned by each of the students in his class. He found that 14 people had no pets, 17 people had one pet, 5 people had two pets and 9 people had three pets.

a Construct a frequency table of the data.

b How many students were surveyed?

c How many pets were owned in total?

d Calculate the mean number of pets owned by Mr Rodriguez's students.

23 Ben asked 35 people about how many siblings they have. He found that 12 people had no siblings, 15 people had one sibling, 3 people had two siblings and 5 people had three siblings.

a Construct a frequency table of Ben's results.

b Find the mode of the data.

c Find the range of the data.

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